FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

General Examination & Growth Assessment

Clinical skill | Week 1 · Breathing, fever & the clinical methods | Dr Mawanda

The general examination of a child starts before you touch them, with observation, then age-appropriate vital signs and accurate anthropometry plotted as WHO Z-scores. Wasting, stunting, MUAC, bilateral oedema, dehydration, pallor and jaundice are detected here and decide treatment. Measuring and plotting growth and interpreting vital signs against age norms are core OSCE skills.

Must know before the ward round

1. **Observe first, touch last**: look at the child on the mother's lap (alertness, breathing, colour, hydration, nutrition) and **count the respiratory rate for a full minute** before disturbing them.
2. **Fast breathing (WHO/IMCI)**: **≥60/min under 2 months; ≥50/min at 2–12 months; ≥40/min at 1–5 years.** Chest indrawing, grunting, nasal flaring and head nodding mean respiratory distress.
3. **Fever** = axillary **≥37.5°C**; **hypothermia <35.5°C** (IMCI). Capillary refill **>3 s** is abnormal (shock = cold hands + CRT >3 s + weak fast pulse).
4. **Hypotension (APLS)**: systolic BP **<70 + (2 × age in years)** mmHg for 1–10 years; **<60** in neonates, **<70** in infants.
5. **Weigh undressed** on a calibrated scale; **length lying until 2 years**, **height standing from 2 years**; head circumference = largest occipitofrontal circumference (3 readings).
6. **Z-score = (measured value − median) ÷ SD** of the WHO reference. **Below −2 = moderate, below −3 = severe.** WFL/WFH → wasting; LFA/HFA → stunting; WFA → underweight.
7. **SAM (6–59 months)**: **WFL/WFH < −3**, or **MUAC <11.5 cm**, or **bilateral pitting oedema**. **MAM**: WFH −3 to −2 or **MUAC 11.5–<12.5 cm**. Infants 6 weeks–6 months: **MUAC <11.0 cm** (WHO 2023).
8. **Bilateral pitting oedema**: press the dorsum of both feet with the thumbs for **3 seconds**: + feet only; ++ feet, legs, hands/forearms; +++ generalised including face. Any bilateral nutritional oedema = SAM.
9. **Dehydration (WHO)**: **severe** = 2 of lethargic/unconscious, sunken eyes, drinks poorly/not able, skin pinch **≥2 s**; **some** = 2 of restless/irritable, sunken eyes, drinks eagerly, skin pinch slow.
10. **Pallor**: compare the child's **palm** with yours (IMCI): **some** palmar pallor → anaemia; **severe** (white palm) → Hb likely <7 g/dL: urgent Hb and assessment for transfusion.
11. **Neonatal jaundice spreads head to toe (Kramer)**: jaundice on the **palms and soles** suggests a high bilirubin; any jaundice **<24 h** of age or on palms/soles needs a measured bilirubin.
12. **Growth landmarks**: birth weight **doubles by 5 months, triples by 1 year**; length **50 → 75 cm** in year 1; head circumference **35 → ~47 cm** in year 1; anterior fontanelle closes **12–18 months**.

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In a nutshell

The general examination is where a child's illness becomes **visible and measurable**. Before you listen to a chest or palpate an abdomen, you should already know whether the child is **sick or well**, **breathing normally or not**, **perfused**, **hydrated**, **pale or jaundiced**, **wasted, stunted or oedematous**, and whether the **vital signs** are normal **for their age**.

Growth is the best single marker of a child's health. A child who is not growing well is telling you something about feeding, chronic infection (HIV, TB), chronic disease or neglect. In Uganda, where **stunting** affects about a quarter of under-fives and **wasting** remains common in some regions, accurate measurement and plotting are not paperwork: they decide who gets **RUTF, F-75, antibiotics and admission**.

CLINICAL PEARL

Examine children **opportunistically**: do what is possible while they are calm (observation, RR, heart and chest), leave upsetting parts for last (temperature with a probe, throat, ears, painful areas, MUAC if they hate it), and use the caregiver's lap and a toy. A screaming child has a falsely high RR and HR and no audible breath sounds.

UGANDA CONTEXT

Uganda uses the **WHO Child Growth Standards** (0–5 years) on the **Child Health Card**, and **MUAC tapes** (red <11.5 cm, yellow 11.5–12.5 cm, green ≥12.5 cm) at every level including VHTs. The **Uganda Clinical Guidelines 2023** and the national **Integrated Management of Acute Malnutrition (IMAM)** guidelines use **WFH/WFL, MUAC, bilateral oedema and the appetite test** to classify malnutrition. **WHO 2023 wasting guideline** adds criteria for infants under 6 months (WFA < −2, WFL < −2, MUAC <11.0 cm from 6 weeks, or oedema).

Equipment

Item	Notes
Infant scale (electronic, 10 g precision)	For children who cannot stand; zero (tare) before use; weigh **naked** or in a light nappy
Hanging (Salter) scale with weighing pants	Field/HC use; read at eye level when still; 100 g precision
Standing scale	Older children, light clothes, no shoes
Length board (infantometer)	**Under 2 years (or <87 cm)**: lying; needs **two people**
Stadiometer (height board)	**2 years and over**: standing, heels, buttocks, shoulders and back of head touching the board
Non-stretch tape measure	Head circumference
MUAC tape (colour-coded)	6–59 months (and adapted cut-offs for older children and infants)
Digital thermometer	Axillary in under-5s (WHO/UCG); tympanic or oral in older children
Pulse oximeter with paediatric/neonatal probes	SpO_2 in every sick child
BP cuffs of several sizes	Bladder width ≈ **40% of arm circumference**, bladder length 80–100% of circumference; too small a cuff over-reads

Item	Notes
Watch with a second hand / timer	RR and HR for a full minute
Stethoscope with paediatric diaphragm, **pen torch**, **otoscope**, tongue depressor, tendon hammer	
WHO growth charts (weight-for-age, length/height-for-age, weight-for-length/height, head circumference-for-age, BMI-for-age 5–19 y), **Z-score tables**	Sex-specific (pink girls, blue boys)
Gloves, hand rub, a toy	Infection control and distraction

Indications & contraindications

- **Every child at every contact** needs a general examination; **weight at every visit**; length/height, MUAC and head circumference (under 2) at least at admission and at growth monitoring.
- **Emergency signs** (ETAT): resuscitate first; complete anthropometry once stable (estimate weight for drug doses meanwhile).
- **Do not stretch a child with a suspected spinal injury or fracture** for length; **do not measure MUAC** on an arm with oedema, a drip or a cast (use the other arm).
- **Respect privacy**: undress only as needed, keep warm (neonates lose heat fast), use a chaperone for genital examination in older children.

Procedure step by step

1. Preparation

1. Hand hygiene; greet the caregiver and child; explain what you will do.
2. Warm hands and stethoscope; warm room for infants.
3. Keep the child on the caregiver's lap where possible; get down to their level.

2. General inspection ("from the end of the bed")

Spend **30–60 seconds just looking**:

- **Sick or well?** Alert and interactive, irritable, lethargic, or unconscious (**AVPU**: Alert, responds to Voice, responds to Pain, Unresponsive; or the **Blantyre coma score** in malaria).
- **Breathing**: rate (count now), effort (**lower chest wall indrawing**, intercostal/subcostal recession, nasal flaring, **grunting**, **head nodding**), noises (stridor, wheeze), ability to talk/feed.
- **Colour**: pallor, central cyanosis (tongue), jaundice, plethora.
- **Hydration**: sunken eyes, dry mouth, tears, irritability or lethargy.
- **Nutrition**: visible severe wasting (loss of buttock fat, "baggy pants", prominent ribs), **oedema** (puffy face, swollen feet), hair changes.
- **Posture and movement**: neck stiffness, opisthotonus, frog-leg posture (hypotonia), asymmetry.
- **Dysmorphic features** (e.g. Down syndrome: upslanting palpebral fissures, epicanthic folds, flat nasal bridge, single palmar crease, sandal gap).
- **Interaction with the caregiver** (warmth, fear, neglect).
- **Equipment around the child**: oxygen, NGT, drip, medicines.

3. Vital signs

Age	Respiratory rate (/min)	Heart rate (/min)	Systolic BP (mmHg)	Approximate weight
Term newborn	25–60	120–170	60–95	3.5 kg
3 months	25–60	115–170	60–105	6 kg
6 months	20–55	110–170	75–105	8 kg
1 year	20–45	105–150	70–105	10 kg
2 years	20–40	95–150	70–105	12 kg
4 years	17–30	80–150	75–110	15 kg
6 years	16–30	75–140	80–115	20 kg
10 years	15–25	60–130	85–120	30 kg
14 years	14–25	60–115	90–125	50 kg

Acceptable ranges (approximately 5th–95th centiles) from the RCH Melbourne clinical guideline. UCG 2023 gives simpler "normal" ranges: <1 year pulse 120–160, RR 30–40, SBP 70–90; 1–5 years pulse 100–120, RR 25–30, SBP 80–90; 6–12 years pulse 80–100, RR 20–25, SBP 90–110.

How to measure each vital sign well

- **Respiratory rate**: count for a **full 60 seconds** with the child calm (abdomen in infants, who breathe diaphragmatically). Use the **WHO fast-breathing cut-offs**: **≥60/min (<2 months), ≥50/min (2–12 months), ≥40/min (1–5 years)**, and in older children >30/min (5–12 years) is fast. Severe tachypnoea or **slow, irregular breathing** in a sick child is ominous.
- **Heart rate**: brachial pulse in infants, radial in older children, or apical auscultation; count a full minute. Tachycardia: fever (about 10/min per °C), pain, anxiety, hypovolaemia, anaemia, heart failure, salbutamol. **Bradycardia in a sick child is a pre-arrest sign** (hypoxia).
- **Capillary refill time**: press on the **sternum** (or fingertip at heart level) for **5 seconds**, release, count: **normal ≤2 s**; **>3 s** with cold hands and a weak fast pulse = **shock** (WHO). Cold room falsely prolongs it.
- **Temperature**: **axillary** for under-5s (digital thermometer, 1–3 minutes or until beep); fever **≥37.5°C** (IMCI) (≥38°C in many international definitions); hypothermia **<35.5°C**; neonates should be **36.5–37.5°C**.
- **SpO_2**: well-perfused finger, toe or (in neonates) the right hand (pre-ductal); wait for a steady waveform. **<90% → oxygen** (WHO); target ≥94% in emergency/ETAT.
- **Blood pressure**: right arm, correct cuff, child calm. **Hypotension** (APLS): SBP **<70 + (2 × age)** for 1–10 years, **<60** neonate, **<70** infant, **<90** over 10 years. BP falls **late** in children: tachycardia and prolonged CRT come first. Measure BP in every child with oedema, haematuria, headache, renal disease or heart disease.
- **Weight and pain score** are often called the "fifth and sixth vital signs" in paediatrics.

4. Anthropometry

Weight

1. Calibrate (zero) the scale. Undress the child (or weigh the mother alone, then mother + baby, and subtract, if only an adult scale is available).
2. Read when still; record to **0.1 kg** (0.01 kg in infants on an electronic scale).
3. **Plot** on the sex-specific chart and compare with previous weights on the card: **the trend matters more than one point**. Newborns lose up to **about 10% of birth weight** in the first days and regain it by **10–14 days**.

Length (under 2 years) / height (2 years and over)

1. **Length**: two people; child lying on the board, head against the fixed headpiece (looking straight up), body straight, knees gently pressed down, feet flat against the movable footpiece; read to **0.1 cm**.
2. **Height**: barefoot, heels together, heels, buttocks, shoulder blades and back of head touching the board, looking straight ahead; lower the headpiece; read to 0.1 cm.
3. If a child ≥2 years is measured lying, **subtract 0.7 cm**; if a child <2 years is measured standing, **add 0.7 cm** (WHO convention).

Head circumference (all children under 2, and any child with neurological concern)

1. Non-stretch tape around the **largest occipitofrontal circumference**: just above the eyebrows and ears, round the most prominent part of the occiput.
2. Take **three readings** and record the **largest**, to 0.1 cm. Plot on the HC-for-age chart.
3. **Microcephaly**: HC **< −2 SD** (severe < −3 SD): congenital infection (CMV, Zika, rubella, toxoplasma), HIE, genetic syndromes, fetal alcohol. **Macrocephaly**: **> +2 SD** or crossing centiles upward: hydrocephalus (post-meningitis, aqueduct stenosis), subdural collection, familial large head.

Mid-upper arm circumference (MUAC), 6–59 months

1. Use the **left arm**, relaxed and hanging by the side.
2. Find the **midpoint between the acromion (shoulder tip) and the olecranon (elbow tip)** with the elbow flexed at 90°; mark it.
3. With the arm relaxed, wrap the tape round the midpoint: **snug, not too tight (pinching) or too loose**; read to **0.1 cm** (1 mm).
4. **Interpret**: **<11.5 cm (red) = SAM**; **11.5–<12.5 cm (yellow) = MAM**; **≥12.5 cm (green) = normal** (12.5–13.5 cm is "at risk" in some programmes). Infants 6 weeks–<6 months (WHO 2023): **<11.0 cm** = at risk of poor growth and development.

5. Z-scores: calculating and interpreting

A **Z-score (standard deviation score)** tells you how many standard deviations a child's measurement is from the **median** of the WHO reference population of the same **sex and age (or length/height)**:

Z = (observed value − median of reference) ÷ standard deviation of reference

Worked example: a 2-year-old boy is 81 cm long. The WHO median length for boys at 24 months is about 87 cm with an SD of about 3.2 cm. Z = (81 − 87) ÷ 3.2 ≈ **−1.9**: borderline, just above the stunting cut-off. In practice you read Z-scores from the **WHO charts or tables** rather than calculate them.

Indicator	What it measures	−2 to −3	Below −3	Above +2
Weight-for-length/height (WFL/WFH)	**Wasting** (acute malnutrition)	**Moderate wasting (MAM)**	**Severe wasting (SAM)**	**Overweight** (> +2), **obese** (> +3) in under-5s
Length/height-for-age (LFA/HFA)	**Stunting** (chronic malnutrition)	**Moderate stunting**	**Severe stunting**	Tall (consider endocrine/genetic causes if extreme)
Weight-for-age (WFA)	**Underweight** (wasting and/or stunting combined)	Underweight	Severely underweight	Use WFH to assess overweight
Head circumference-for-age	Brain growth	Microcephaly (< −2)	Severe microcephaly	Macrocephaly
BMI-for-age (5–19 years)	Thinness/overweight	Thinness (< −2)	Severe thinness (< −3)	Overweight (> +1), obesity (> +2)

Classifying acute malnutrition (6–59 months; WHO 2023 / UCG 2023)

Classification	Criteria
Severe acute malnutrition (SAM) / severe wasting	**WFH/WFL < −3 Z**, **or MUAC <11.5 cm**, **or bilateral pitting oedema** (any grade)
Complicated SAM (inpatient)	SAM **plus** any medical complication (danger signs, severe pneumonia, severe dehydration, hypothermia, hypoglycaemia, severe anaemia, high fever, oedema +++), **or failed appetite test** (cannot finish RUTF test portion), or age <6 months
Uncomplicated SAM (outpatient therapeutic care, OTC)	SAM, passes the appetite test, no complications
Moderate acute malnutrition (MAM)	**WFH −3 to < −2 Z** or **MUAC 11.5–<12.5 cm**, no oedema

Reading a growth chart: plot several points; **parallel to the centile/Z-lines** = normal growth; **crossing two major lines downward** (e.g. from 0 to −2) = **growth faltering**; a flat line = no weight gain (illness, feeding problem, neglect). **Upward crossing** after treatment = catch-up growth.

Mid-parental height (target height, older children with short stature): boys = (father's height + mother's height + 13 cm) ÷ 2; girls = (father + mother − 13 cm) ÷ 2; ±8.5 cm range.

6. Head-to-toe general examination

Area	What to look for	Significance
Head	Size and shape; **anterior fontanelle** (size, tension: **bulging** in a calm upright infant = raised ICP/meningitis; **sunken** = dehydration); sutures; **hair** (sparse, pale, easily plucked, "flag sign" in kwashiorkor)	Meningitis, hydrocephalus, dehydration, malnutrition
Face	Dysmorphism, facial oedema, parotid swelling (HIV, mumps), facial palsy	Syndromes, nephrotic syndrome, HIV
Eyes	**Conjunctival pallor**, **scleral jaundice**, sunken eyes, **Bitot spots / xerosis / corneal clouding or ulcer** (vitamin A deficiency), conjunctivitis (measles), periorbital oedema, squint	Anaemia, jaundice, dehydration, xerophthalmia (urgent vitamin A)
Mouth	**Oral thrush**, ulcers (measles, herpes, SJS), dry mucosa, **dentition** (absence of canines: ebiino extraction), gum bleeding, tonsils (last)	HIV, dehydration, vitamin C deficiency, mucositis
Ears	Discharge, tender mastoid (otoscopy last)	Chronic otitis media (HIV), mastoiditis
Neck	**Lymph nodes**, neck stiffness (meningism), thyroid, torticollis	TB adenitis, meningitis
Hands	**Palmar pallor** (compare with your palm), **finger clubbing** (cyanotic heart disease, bronchiectasis, chronic suppurative lung disease from HIV/TB, IBD), peripheral cyanosis, cold peripheries, nail changes (koilonychia), **dactylitis** (swollen painful fingers: sickle cell in infants), palmar crease	Anaemia, chronic hypoxia, shock, SCD
Skin	**Rash** (maculopapular measles rash, petechiae/purpura = meningococcaemia, ITP, leukaemia), **skin turgor** (pinch the abdomen halfway between umbilicus and flank for 1 s), **kwashiorkor dermatosis** ("flaky paint"), scabies, impetigo, **BCG scar** (right upper arm), burns/bruises (pattern? child protection), café-au-lait spots	Infection, dehydration, malnutrition, abuse
Lymph nodes	Cervical, axillary, inguinal, epitrochlear: size, tenderness, consistency, matting	Small (<1 cm cervical/axillary, <1.5 cm inguinal) mobile nodes are common; **generalised lymphadenopathy** (≥2 non-contiguous sites) → HIV, TB, leukaemia; matted cervical nodes → TB

Area	What to look for	Significance
Feet and legs	**Bilateral pitting oedema** (press the dorsum of both feet firmly with your thumbs for **3 seconds**: a pit that remains = oedema), bowing of legs (rickets), clubfoot	SAM (kwashiorkor), nephrotic syndrome, heart failure
Spine and gait	Scoliosis, gibbus (TB spine), sacral dimple/tuft, limp	Pott's disease, spina bifida occulta
Genitalia (with consent, chaperone)	Undescended testes, hernia, hydrocele, hypospadias, ambiguous genitalia, labial adhesions; **pubertal stage** (Tanner) in older children	Surgical referral, endocrine disorders, abuse signs

7. Specific general signs: how to grade them

Oedema (nutritional; WHO/IMAM grading)

Grade	Distribution
+ (mild)	Both feet (ankles)
++ (moderate)	Both feet + lower legs, **or** + hands/lower arms
+++ (severe)	Generalised: feet, legs, hands, arms **and face** (periorbital)

Oedema must be **bilateral and pitting** to count for SAM. Unilateral swelling suggests a local cause (cellulitis, DVT, lymphatic obstruction). **Other causes of bilateral oedema**: **nephrotic syndrome** (periorbital first, heavy proteinuria), **heart failure** (hepatomegaly, raised JVP in older children, crackles), liver failure, severe anaemia, protein-losing enteropathy, fluid overload.

Dehydration (WHO; UCG 2023)

Sign	No dehydration	Some dehydration	Severe dehydration
General condition	Well, alert	**Restless, irritable**	**Lethargic or unconscious**
Eyes	Normal	**Sunken**	**Sunken**
Drinking	Normally	**Eagerly, thirsty**	**Poorly or not able**
Skin pinch	Goes back immediately	**Slowly** (<2 s)	**Very slowly (≥2 s)**
Plan	**A**	**B**	**C**

Two or more signs place the child in a category. **Caution in SAM**: sunken eyes and slow skin pinch are present in wasted children without dehydration; rely on history of fluid loss, recent change in eyes (ask the mother) and thirst, and treat with ReSoMal cautiously.

Pallor (IMCI)

- Compare the child's **palm** with your own and the mother's: **some palmar pallor** (paler than normal) or **severe palmar pallor** (very pale, almost white).
- Supplement with **conjunctivae, tongue and nail beds**.
- Clinical pallor is a **screening tool**: severe palmar pallor detects most children with Hb <7 g/dL but misses some; **measure Hb** (HemoCue) in any pale child.
- Look for associated **signs**: tachycardia, flow murmur, **gallop rhythm, hepatomegaly and respiratory distress** (decompensation: urgent transfusion), jaundice and splenomegaly (haemolysis: malaria, sickle cell), petechiae (marrow failure, leukaemia), koilonychia (iron deficiency).

Jaundice

- Examine in **daylight**; **blanch the skin** with a finger (forehead, sternum, knees, soles) and look at the colour underneath; check sclerae and gums.
- **Neonatal jaundice progresses cephalocaudally (Kramer zones)**: 1 head and neck; 2 upper trunk; 3 lower trunk and thighs; 4 arms and lower legs; 5 **palms and soles**. Higher zones mean higher bilirubin (zone 5 roughly >250 µmol/L) but visual estimation is **unreliable**, especially in dark skin: **measure** (TcB/TSB) whenever in doubt.
- **Danger features**: jaundice in the **first 24 hours**, on the **palms and soles**, in a **preterm or sick** baby, **pale stools/dark urine** (conjugated), or **signs of bilirubin encephalopathy** (lethargy, poor feeding, high-pitched cry, hypertonia, arching, setting-sun eyes).
- Older children: jaundice with fever → malaria (haemolysis), hepatitis, sepsis; with pallor and splenomegaly → haemolysis (sickle cell, G6PD); with pale stools → obstruction.

Signs of HIV on general examination: oral thrush (beyond the neonatal period or recurrent), **generalised lymphadenopathy**, **parotid enlargement**, hepatosplenomegaly, chronic ear discharge, finger clubbing, failure to thrive, recurrent skin infections, herpes zoster in a child.

Signs of vitamin and mineral deficiency: **Bitot spots/xerophthalmia** (vitamin A), **rickets** (widened wrists, rachitic rosary, bowed legs, frontal bossing, delayed fontanelle closure), **angular stomatitis** (B vitamins, iron), **koilonychia** (iron), **goitre** (iodine), gum bleeding (vitamin C).

OSCE checklist

Step	Done?
Hand hygiene, introduces self, consent from caregiver (and assent from older child)	
Positions the child comfortably (caregiver's lap), uses distraction	
General inspection: sick/well, AVPU, breathing effort, colour, hydration, nutrition, dysmorphism	
Counts RR for a full minute before disturbing the child; interprets with age cut-offs	
Measures HR (full minute), CRT, temperature (axillary in under-5s), SpO_2, BP with correct cuff	
Weighs correctly (calibrated scale, undressed) and records to 0.1 kg	
Measures length (lying, <2 y, two people) or height (standing, ≥2 y) correctly	
Measures head circumference (largest of 3) in a child under 2	
Measures MUAC correctly (left arm, midpoint, snug) and states the colour/category	
Plots on the correct sex-specific WHO chart and states Z-scores (WFH, HFA, WFA)	
Checks bilateral pitting oedema on the feet (3 s) and grades it	
Assesses dehydration (eyes, drinking, skin pinch, general condition) and classifies	
Checks palmar and conjunctival pallor; jaundice in daylight with blanching	
Examines head (fontanelle), eyes, mouth (thrush), lymph nodes, hands (clubbing), skin (rash, BCG scar)	
Thanks the family, covers the child, washes hands	
Summarises findings and nutritional classification and states the next step	

Clinical acumen

Focused history that guides the examination

- **Feeding and weight trend** (card): poor feeding history → look carefully for wasting, oedema, micronutrient signs.
- **Diarrhoea and vomiting** → dehydration signs; with SAM, interpret them cautiously.
- **Fever** → rash, neck stiffness, ears, throat, lymph nodes, joints, jaundice and pallor (malaria).
- **Swelling** → where it started (eyes first: nephrotic; feet first: kwashiorkor or heart failure), urine (frothy, red), breathlessness.
- **HIV exposure** → oral thrush, lymphadenopathy, parotids, growth.
- **Birth history** → head circumference, tone, dysmorphism.

Examination: making the measurements trustworthy

- **Weigh undressed** and on the **same scale**; clothes and nappies can add 0.3–0.5 kg (10% of a small infant's weight).
- **Length needs two people**; a wriggling child measured by one person is often 1–3 cm "shorter", which can falsely label stunting.
- **MUAC** depends on the midpoint and tension: practise on colleagues; repeat if near a cut-off.
- **RR** must be counted, not estimated; a crying child cannot be classified.
- **Use the correct chart**: sex-specific, WFL for <2 years (or <87 cm), WFH for ≥2 years.

Red flags

RED FLAG

- **Emergency signs**: obstructed breathing, central cyanosis, severe respiratory distress, **shock** (cold hands + CRT >3 s + weak fast pulse), coma, convulsions, **severe dehydration** in a lethargic child.
- **Bradycardia or slow irregular breathing** in a sick child: imminent arrest.
- **Hypothermia (<35.5°C)** in a neonate or malnourished child: sepsis and hypoglycaemia risk.
- **SAM with any complication**, oedema +++, or failed appetite test: inpatient care.
- **Severe palmar pallor** with breathlessness, gallop rhythm or hepatomegaly: urgent transfusion.
- **Jaundice in the first 24 h**, on palms and soles, or with neurological signs.
- **Bulging fontanelle** in a calm febrile infant: meningitis.
- **Non-blanching rash** (petechiae/purpura) with fever: meningococcal sepsis until proven otherwise.
- **Rapidly enlarging head** or setting-sun eyes: hydrocephalus.
- **Injuries inconsistent with the history**, patterned bruises or burns: child protection.

Thinking like a clinician

- **"Is this vital sign abnormal for this age?"** RR 44 is normal for a 4-month-old and fast for a 3-year-old.
- **"Is the abnormality explained by fever, crying or fear?"** Recount when calm and afebrile.
- **"Which Z-score tells me what?"** Wasting = acute (treat now), stunting = chronic (prevent further loss, investigate chronic disease), underweight = either.
- **"Does the oedema mean kwashiorkor or nephrotic syndrome?"** Check urine protein, BP, skin and hair changes, diet history and age (nephrotic peaks at 2–6 years; kwashiorkor 1–3 years with poor diet).
- **"Are the dehydration signs reliable here?"** In SAM, no.

- **Common student mistakes**: weighing clothed children; measuring a 1-year-old standing; plotting on the wrong sex chart; using WFA alone to diagnose SAM; missing oedema because the feet were not pressed for 3 s; calling a malnourished child "severely dehydrated" and giving Plan C; estimating the RR; checking only conjunctival pallor; judging neonatal jaundice in poor light.

Clinical workup

Step 0: First 5 minutes

1. **ETAT triage**: look for emergency signs (ABCD: airway/breathing, circulation, coma, convulsions, dehydration) and **treat immediately**.
2. If stable, check **priority signs** (the "3 TPR MOB": **T**iny baby, **T**emperature very high, **T**rauma, **P**allor (severe), **P**oisoning, **P**ain (severe), **R**espiratory distress, **R**estless/irritable/lethargic, **R**eferral (urgent), **M**alnutrition (visible severe wasting), **O**edema of both feet, **B**urns).
3. Count the **RR** and check **SpO_2, CRT, temperature** early; check **glucose** if lethargic or convulsing.

Step 1: Focused history

- [ ] Age (months), sex, **previous weights** on the card (trend).
- [ ] Feeding history (to interpret wasting/stunting).
- [ ] Fluid losses (diarrhoea, vomiting) and urine output (dehydration).
- [ ] Onset and spread of swelling (oedema).
- [ ] Birth weight and gestation (growth expectations; HC).
- [ ] HIV exposure and TB contact (growth faltering).

Step 2: Focused examination

- **General inspection**, then **vital signs**, then **anthropometry** (weight, length/height, HC, MUAC), then **general signs** (oedema, dehydration, pallor, jaundice, lymph nodes, skin, mouth, hands), then the **systems** suggested by the history.
- Document **key negatives**: "no oedema, no pallor, no jaundice, no lymphadenopathy, no thrush".

Step 3: Bedside tests

Test	When	Interpretation
Plotting growth (WFL/WFH, LFA/HFA, WFA, HC)	Every admission and growth visit	Z-score categories; trend
MUAC	6–59 months; adapted cut-offs for older children	SAM <11.5 cm, MAM 11.5–<12.5 cm
Appetite test (RUTF)	SAM 6–59 months	Fails = complicated SAM, admit
Blood glucose	Lethargy, convulsions, SAM, hypothermia, neonates	Treat <3.0 mmol/L in SAM (UCG)
HemoCue Hb	Pallor	Severe anaemia thresholds
Urine dipstick	Oedema (protein), haematuria, hypertension	3+ protein → nephrotic syndrome
Malaria RDT	Fever, pallor, jaundice	Treat malaria
Transcutaneous bilirubin	Neonatal jaundice	Confirm with serum bilirubin near threshold

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
FBC and film	Pallor, lymphadenopathy, hepatosplenomegaly, petechiae	Anaemia type, blasts, low platelets	Transfusion, iron, oncology referral (HC IV and above)
Serum albumin, urine protein:creatinine	Bilateral oedema	Albumin <25 g/L with heavy proteinuria → nephrotic; low albumin without proteinuria → kwashiorkor/protein loss	Steroids vs nutritional rehabilitation (regional referral)
Electrolytes, glucose	Severe dehydration, SAM with complications	Hypokalaemia, hyponatraemia, hypoglycaemia	Fluid choice, potassium (regional referral)
HIV test (PCR <18 months)	Growth faltering, lymphadenopathy, thrush, parotid enlargement	Positive	ART, cotrimoxazole (HC III and above via hub)
TB investigations	Weight loss, lymphadenopathy, contact	MTB detected / CXR changes	TB treatment
Serum bilirubin (total and conjugated)	Neonatal jaundice near threshold, jaundice <24 h, prolonged jaundice	Above phototherapy line; conjugated fraction high	Phototherapy, exchange, biliary atresia work-up
Thyroid function, bone age, karyotype	Short stature without obvious cause, dysmorphism	Hypothyroidism, Turner syndrome	Hormone therapy, genetics (national referral)
Skull X-ray / cranial ultrasound / CT	Abnormal head growth or shape	Hydrocephalus, craniosynostosis	Neurosurgery (national referral; CURE Children's Hospital Mbale for hydrocephalus)
Wrist X-ray, calcium, phosphate, ALP	Signs of rickets	Cupping/fraying, high ALP, low phosphate	Vitamin D and calcium

Step 5: Putting it together

Model summary of a general examination: "A **14-month-old girl**, **6.4 kg**, length **71 cm**: **WFL Z-score below −3**, **LFA −2.4**, **WFA below −3**; **MUAC 11.0 cm** (red); **no oedema**. She is alert but irritable, RR 38/min, HR 148/min, CRT 2 s, temperature 36.8°C axillary, SpO_2 97%. Eyes not sunken by the mother's account, drinks eagerly, skin pinch slow (but she is wasted). **Some palmar pallor**, no jaundice, **oral thrush**, **generalised lymphadenopathy** (cervical, axillary, inguinal 1–1.5 cm), BCG scar present. **Impression**: **severe acute malnutrition (severe wasting) with moderate stunting**, possible HIV infection, anaemia; dehydration status uncertain because of SAM. **Next**: appetite test, glucose, Hb, HIV test (DNA PCR if <18 months), malaria test; manage per SAM protocol."

Worked example: plotting and interpreting A boy aged 9 months weighs 6.9 kg, length 69.5 cm.

- **WFA**: WHO median at 9 months ≈ 8.9 kg; 6.9 kg lies **between −2 and −3 → underweight**.
- **LFA**: median ≈ 72 cm; 69.5 cm lies **between −1 and −2** → not stunted.
- **WFL**: median weight for 69.5 cm ≈ 8.3 kg; 6.9 kg lies **between −2 and −3 → moderate wasting (MAM)**.
- **MUAC 12.1 cm** → yellow → **MAM**. No oedema.
- **Conclusion**: moderate acute malnutrition; counsel on feeding, check for illness (HIV, TB, diarrhoea), supplementary feeding where available, follow-up every 2 weeks.

Worked example: oedematous child A 2-year-old has swelling of both feet and legs and hands for 2 weeks, sparse pale hair, peeling skin on the thighs, MUAC 12.8 cm, WFH −1.5. Urine protein **trace**. **Interpretation**: **bilateral pitting oedema ++ = SAM (kwashiorkor) even with a "normal" MUAC and WFH**; nephrotic syndrome is unlikely with trace proteinuria. Manage as SAM (oedema ++ with complications → inpatient; F-75 100 mL/kg/day if +++).

Management

"Management" for the general examination means **acting on what you find**: the general signs and growth measurements trigger specific treatment pathways.

1. Abnormal vital signs

1. **Fast breathing / indrawing / SpO_2 <90%** → oxygen, treat pneumonia or wheeze per protocol; reassess RR after treatment.
2. **Shock (cold hands, CRT >3 s, weak fast pulse)** → IV/IO access, **fluid bolus** (non-SAM: Ringer's lactate 10–20 mL/kg, UCG says 20 mL/kg rapidly; **SAM: 15 mL/kg over 1 h** of a dextrose-containing fluid; **severe anaemia: blood**), glucose, reassess.
3. **Fever** → find the focus; antipyretic for discomfort (paracetamol 15 mg/kg); malaria test.
4. **Hypothermia (<35.5°C)** → skin-to-skin (kangaroo) care, warm room and blankets, cover the head, check glucose, treat sepsis; in SAM, feed and warm.
5. **Hypertension** (above the 95th centile for age, sex and height) → repeat with the correct cuff; check urine (glomerulonephritis), renal function; refer.

2. Malnutrition pathways (UCG 2023 / WHO 2023 / Uganda IMAM)

Finding	Action
SAM with complications or failed appetite test, or any SAM under 6 months	**Admit to inpatient therapeutic care** (stabilisation): treat/prevent hypoglycaemia, hypothermia, dehydration (ReSoMal), infection (antibiotics), start **F-75**; correct micronutrients; then transition to F-100/RUTF
Uncomplicated SAM (6–59 months, passes appetite test)	**Outpatient therapeutic care (OTC)**: **RUTF 150–185 kcal/kg/day** (WHO 2023), **amoxicillin** (UCG), weekly/fortnightly follow-up, treat malaria, deworm, vaccinate, HIV test
MAM	Nutrition counselling, **supplementary feeding** (specially formulated foods where available, prioritised for MUAC 11.5–11.9 cm, WAZ < −3, age <24 months: WHO 2023), treat illnesses, follow-up every 2 weeks
Infant <6 months at risk of poor growth and development (WHO 2023)	Breastfeeding support (re-lactation), assess the mother's health, treat illness; inpatient if severe or medical complications
Stunting	Counsel on diet quality, hygiene, breastfeeding, and look for chronic disease (HIV, TB, chronic diarrhoea, heart disease); monitor height velocity; stunting is largely **prevented**, not "treated", after 2 years
Overweight/obesity	Diet and activity counselling for the family; screen BP and, if severe, glucose

Discharge from SAM treatment (WHO 2023): WFH/WFL ≥ −2 Z or MUAC ≥12.5 cm for **two consecutive visits**, no oedema for 2 weeks, clinically well.

3. Dehydration

Treat by **Plan A, B or C** (UCG 2023) according to the classification, with **zinc** and continued feeding; **SAM → ReSoMal** (5 mL/kg every 30 min for 2 h, then 5–10 mL/kg/h alternating with F-75). Reassess signs every 1–2 hours (eyes, thirst, skin pinch, urine, consciousness) and stop when rehydrated.

4. Pallor and anaemia

1. **Measure Hb**; test for malaria.
2. **Severe anaemia** (Hb <4 g/dL, or 4–6 with distress, shock, impaired consciousness, heart failure, or high parasitaemia) → **transfuse** (20 mL/kg whole blood or 10 mL/kg packed cells; SAM 10 mL/kg over 3 h).
3. Moderate anaemia → treat cause (malaria, worms, iron deficiency): iron 3 mg/kg/day elemental, albendazole, dietary advice; review Hb in 4 weeks.

5. Jaundice

- **Neonate**: measure bilirubin and plot by hours of age (AAP 2022 nomograms); **phototherapy** or **exchange transfusion** as indicated; encourage frequent breastfeeding; find the cause (ABO/Rh, G6PD, sepsis).
- **Older child**: treat the cause (malaria, hepatitis, haemolysis in SCD); refer obstructive jaundice.

6. Abnormal head size

Microcephaly: developmental assessment, congenital infection screen, hearing and vision; **macrocephaly/rapid growth**: cranial ultrasound (open fontanelle) or CT, neurosurgical referral (hydrocephalus shunt or endoscopic third ventriculostomy: CURE Children's Hospital Mbale is the national centre for paediatric hydrocephalus).

7. Communicating growth findings

Show the caregiver the chart: "This line is where most healthy girls her age are. Her weight was here 3 months ago and has gone down to here. We need to find out why and help her gain weight." Avoid blaming; focus on a feeding plan the family can afford; book follow-up.

Complications

- **Errors in measurement** leading to wrong classification (missed SAM, unnecessary admission).
- **Hypothermia** in undressed neonates and malnourished children during examination.
- **Distress and loss of cooperation** if painful procedures come first.
- **Cross-infection** (scales, MUAC tapes, stethoscopes: clean between children).
- **Over-treatment of "dehydration" in SAM** (heart failure).
- **Missed child protection** if the skin is not fully examined.

Ward-round & exam pearls

Model case presentation (focus on general examination and growth): "**Atim**, a **2-year-and-1-month-old girl**, is **fully immunised**, **HIV-unexposed**, and was **breastfed until 18 months**. She weighs **8.2 kg**, height **79 cm**: **WFH Z-score −2.6**, **HFA −2.9**, **WFA −3.4**; **MUAC 12.0 cm**; **no oedema**. She is alert, RR 32/min, HR 128/min, CRT 2 s, temperature 37.9°C axillary, SpO_2 98%, BP 90/55. She has **some palmar pallor**, no jaundice, no lymphadenopathy, no thrush; **BCG scar** present; skin and hair normal. **Summary**: **moderate wasting (MAM) with moderate stunting** and mild anaemia in a febrile child. **Plan**: find the cause of fever (RDT, urine), Hb, feeding counselling with a 24-hour recall, supplementary feeding, deworm, review in 2 weeks."

Viva questions with answers

1. *"What are the WHO cut-offs for fast breathing?"* ≥60/min under 2 months, ≥50/min at 2–12 months, ≥40/min at 1–5 years.
2. *"How do you diagnose SAM?"* In 6–59 months: WFH/WFL < −3 Z, MUAC <11.5 cm, or bilateral pitting oedema; complicated if danger signs/complications or a failed appetite test.
3. *"Why measure length lying before 2 years?"* Infants cannot stand straight reliably; WHO standards use recumbent length until 24 months (adjust by 0.7 cm if the other method is used).
4. *"How do you check for nutritional oedema?"* Press the dorsum of both feet with the thumbs for 3 seconds; a pit on both feet is oedema; grade +, ++, +++.
5. *"What are the signs of severe dehydration?"* Two of: lethargic/unconscious, sunken eyes, unable to drink or drinks poorly, skin pinch goes back very slowly (≥2 s).
6. *"Where do you measure MUAC?"* Left arm, midpoint between acromion and olecranon, arm relaxed, tape snug.
7. *"What is a Z-score?"* (Measured value − reference median) ÷ reference SD for the same sex and age/length.
8. *"How does neonatal jaundice spread and why is that useful?"* Head to toe (Kramer zones); palms and soles suggest a high level, but always measure.

OSCE tips

- Say what you are doing out loud: "I am counting the respiratory rate for a full minute while she is calm on her mother's lap."
- **State the age-specific interpretation** of every vital sign.
- Always **plot**, then **name the Z-score category**, then **give the nutritional diagnosis**.
- Do not forget **oedema** before you classify nutrition; it overrides the numbers.
- Keep the child warm and comfortable; examiners mark rapport.

Mnemonics & memory aids

- **Fast breathing "60-50-40"**: under 2 months, 2–12 months, 1–5 years.
- **Hypotension "70 + 2 × age"**.
- **Shock "Cold, Slow, Weak, Fast"**: cold hands, slow capillary refill (>3 s), weak and fast pulse.
- **SAM "3-11.5-oedema"**: WFH below −3, MUAC below 11.5 cm, bilateral oedema.
- **MUAC colours "Red-Yellow-Green" = "Stop-Caution-Go"**: <11.5, 11.5–12.5, ≥12.5 cm.
- **Z-scores "2 is a warning, 3 is an emergency"** (below −2 moderate, below −3 severe).
- **Oedema grades "Feet-Legs/Arms-Face"**: +, ++, +++.
- **Growth "Double at 5 months, triple at 1 year"**; length "50 → 75 → 100" at birth, 1 year and 4 years; head "35 → 47" in year 1.
- **Priority signs "3 TPR MOB"**: Tiny baby, Temperature, Trauma; Pallor, Poisoning, Pain; Respiratory distress, Restless/lethargic, Referral; Malnutrition, Oedema of both feet, Burns.

High-yield summary

Numbers & doses to know

Item	Value
Fast breathing	**≥60** (<2 m), **≥50** (2–12 m), **≥40** (1–5 y)
Fever / hypothermia	Axillary **≥37.5°C** / **<35.5°C**; neonate normal 36.5–37.5°C
Capillary refill	Normal **≤2 s**; shock **>3 s** with cold hands and weak fast pulse

Item	Value
Hypotension	SBP **<70 + 2 × age** (1–10 y); <60 neonate; <70 infant; <90 >10 y
Oxygen threshold	SpO_2 **<90%** (WHO)
SAM (6–59 m)	**WFH < −3**, **MUAC <11.5 cm**, or **bilateral oedema**
MAM	WFH −3 to < −2, **MUAC 11.5–<12.5 cm**
Infant 6 weeks–6 months at risk (WHO 2023)	WFA < −2, WFL < −2, **MUAC <11.0 cm**, or oedema
Stunting	**HFA < −2** (severe < −3)
Overweight / obesity (under 5)	WFH > +2 / > +3
Microcephaly	HC < −2 SD (severe < −3)
Length vs height	Lying **<2 years**, standing **≥2 years**; adjust **0.7 cm**
Birth weight	Doubles by **5 months**, triples by **1 year**, quadruples by **2 years**
Length	**50 cm** birth, **75 cm** 1 year, ~**87 cm** 2 years, ~**100 cm** 4 years
Head circumference	**35 cm** at birth, ~**47 cm** at 1 year (+12 cm in year 1)
Neonatal weight loss	Up to ~**10%**; regained by **10–14 days**
Anterior fontanelle	Closes **12–18 months**; posterior by ~2 months
Skin pinch	Severe dehydration **≥2 s**
SAM discharge (WHO 2023)	WFH ≥ −2 or MUAC ≥12.5 cm on 2 consecutive visits, no oedema

Classic exam traps

- **Trap:** using weight-for-age alone to diagnose SAM. **Right:** WFH/WFL, MUAC and oedema; WFA mixes wasting and stunting.
- **Trap:** a child with bilateral oedema and a normal MUAC is "not malnourished". **Right:** bilateral pitting oedema = SAM (after excluding nephrotic syndrome and heart failure).
- **Trap:** classifying a wasted child with sunken eyes and slow skin pinch as "severe dehydration" and giving Plan C. **Right:** these signs are unreliable in SAM; use history, ReSoMal and careful monitoring.
- **Trap:** RR 46 in a 3-month-old is "fast breathing". **Right:** the cut-off at 2–12 months is ≥50.
- **Trap:** measuring a 15-month-old standing. **Right:** lying length under 2 years.
- **Trap:** assessing neonatal jaundice in a dim ward and calling it mild. **Right:** daylight, blanch the skin, and measure bilirubin.
- **Trap:** BP is normal, so the child is not in shock. **Right:** hypotension is a late sign; tachycardia, prolonged CRT and cold hands come first.

Questions seniors ask

1. **"What are her Z-scores and what do they mean?"** Give WFH (wasting), HFA (stunting), WFA (underweight) and MUAC, then the classification.
2. **"Is that respiratory rate fast for her age?"** Compare with 60/50/40.
3. **"Does this child have shock?"** Cold hands, CRT >3 s, weak fast pulse; BP is late.
4. **"Show me how you check for oedema."** Both feet, thumbs, 3 seconds, look for pits; grade.
5. **"How would you tell kwashiorkor from nephrotic syndrome?"** Diet history, skin and hair changes, urine protein (heavy in nephrotic), albumin, age, periorbital onset in nephrotic.
6. **"Is she dehydrated?"** Use the WHO signs; in SAM, be cautious and rely on history and response to ReSoMal.

7. **"How much should a 1-year-old weigh?"** About three times birth weight (≈10 kg), plotted on the chart rather than guessed.
8. **"How will you explain the growth chart to the mother?"** Show the curve and her child's points, explain the trend, agree a feeding plan and follow-up without blame.

SELF-TEST

Flashcards

33 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	WHO fast-breathing cut-offs	**≥60/min** under 2 months; **≥50/min** at 2–12 months; **≥40/min** at 1–5 years. Count for a full minute with the child calm.
2	Signs of respiratory distress in a child	**Lower chest wall indrawing**, nasal flaring, **grunting**, head nodding, tracheal tug, inability to feed or talk, cyanosis.
3	WHO definition of shock in a child	**Cold hands + capillary refill >3 s + weak, fast pulse** (all three).
4	How do you measure capillary refill?	Press on the **sternum** (or a fingertip at heart level) for **5 s**, release and count; normal **≤2 s**. A cold room prolongs it.
5	APLS hypotension formula	Systolic BP **<70 + (2 × age in years)** for 1–10 years; <60 neonates; <70 infants; <90 over 10 years. Hypotension is a **late** sign.
6	Fever and hypothermia thresholds (IMCI)	Fever: axillary **≥37.5°C**. Hypothermia: **<35.5°C**. Neonatal normal 36.5–37.5°C.
7	Normal heart rate ranges (approx.)	Newborn 120–170; 1 year 105–150; 2 years 95–150; 6 years 75–140; 10 years 60–130 /min (RCH).
8	Length or height?	**Length lying under 2 years** (two people, infantometer); **height standing from 2 years**. Adjust by **0.7 cm** if the other method is used.
9	How to measure head circumference	Non-stretch tape over the **largest occipitofrontal circumference** (above eyebrows and ears, round the occiput); **3 readings**, record the largest.
10	How to measure MUAC	**Left arm**, **midpoint between acromion and olecranon** (elbow at 90° to find it), arm relaxed, tape snug (not pinching); read to 1 mm.
11	MUAC cut-offs (6–59 months)	**<11.5 cm (red) = SAM**; **11.5–<12.5 cm (yellow) = MAM**; **≥12.5 cm (green) = normal**. Infants 6 weeks–6 months: **<11.0 cm** at risk (WHO 2023).
12	Z-score formula	**Z = (measured value − reference median) ÷ reference SD** for the same sex and age (or length/height).
13	What does each growth indicator measure?	**WFH/WFL → wasting** (acute); **HFA/LFA → stunting** (chronic); **WFA → underweight** (mixed); HC-for-age → brain growth; BMI-for-age (5–19 y) → thinness/obesity.
14	Z-score interpretation	Below **−2 = moderate**, below **−3 = severe** (wasting, stunting, underweight). WFH above +2 = overweight, above +3 = obese (under 5).
15	SAM criteria (6–59 months)	**WFH/WFL < −3 Z**, or **MUAC <11.5 cm**, or **bilateral pitting oedema**.
16	Complicated SAM	SAM **plus** a medical complication (danger signs, severe pneumonia, dehydration, hypothermia, hypoglycaemia, severe anaemia, oedema +++) **or failed RUTF appetite test**, or age <6 months → inpatient care.
17	How to test for nutritional oedema	Press the **dorsum of both feet** with the thumbs for **3 seconds**; a pit remaining on **both** feet = bilateral pitting oedema.
18	Oedema grades	**+** both feet; **++** feet + lower legs, or + hands/forearms; **+++** generalised including the **face**.
19	WHO signs of severe dehydration	Two of: **lethargic/unconscious**, **sunken eyes**, **drinks poorly or unable**, **skin pinch goes back very slowly (≥2 s)**.

#	Question	Answer
20	WHO signs of some dehydration	Two of: **restless/irritable**, **sunken eyes**, **drinks eagerly/thirsty**, **skin pinch goes back slowly**.
21	Why are dehydration signs unreliable in SAM?	Wasting itself causes **sunken eyes and loss of skin turgor**; rely on history of losses, recent change, thirst; treat with **ReSoMal** cautiously.
22	How to assess palmar pallor (IMCI)	Compare the child's **palm** with your own: **some pallor** (paler) or **severe pallor** (very pale/white). Confirm with **Hb**.
23	Kramer zones of neonatal jaundice	1 head/neck; 2 upper trunk; 3 lower trunk/thighs; 4 arms/lower legs; 5 **palms and soles** (high bilirubin). Visual estimates are unreliable: measure.
24	Danger features of neonatal jaundice	Onset **<24 h**, **palms/soles**, preterm or sick baby, **pale stools/dark urine**, lethargy/poor feeding/high-pitched cry/arching (encephalopathy).
25	Growth landmarks: weight	Birth weight **doubles by 5 months**, **triples by 1 year**, quadruples by 2 years; newborns lose up to ~10% and regain by **10–14 days**.
26	Growth landmarks: length and head	Length **50 cm** at birth, **75 cm** at 1 year, ~**100 cm** at 4 years. HC **35 cm** at birth, ~**47 cm** at 1 year.
27	Fontanelle closure	Posterior by about **2 months**; anterior **12–18 months**. **Bulging** (calm, upright) → raised ICP/meningitis; **sunken** → dehydration.
28	Signs of HIV on general examination	Oral thrush, **generalised lymphadenopathy**, **parotid enlargement**, hepatosplenomegaly, chronic ear discharge, clubbing, failure to thrive, recurrent skin infections.
29	Kwashiorkor vs nephrotic syndrome	Kwashiorkor: poor diet, **skin (flaky paint) and hair changes**, wasting, little proteinuria. Nephrotic: **periorbital onset**, **heavy proteinuria (3+)**, low albumin, high cholesterol, usually well-nourished.
30	Mid-parental height	Boys: (father + mother + 13 cm) ÷ 2; girls: (father + mother − 13 cm) ÷ 2; target range ±8.5 cm.
31	BP cuff size	Bladder width about **40% of arm circumference**, length **80–100%** of circumference; too small a cuff **over-reads**.
32	SAM discharge criteria (WHO 2023)	WFH/WFL **≥ −2** or MUAC **≥12.5 cm** on **two consecutive visits**, no oedema for 2 weeks, clinically well.
33	Priority signs (ETAT) mnemonic	**3 TPR MOB**: Tiny baby, Temperature very high, Trauma; Pallor, Poisoning, Pain; Respiratory distress, Restless/lethargic, Referral; Malnutrition, Oedema of both feet, Burns.

SELF-TEST

Single best answer questions

16 questions. The answers and explanations follow at the end.

1. A 4-month-old boy with cough has a respiratory rate of 46/min counted over a full minute while calm, no chest indrawing, SpO_2 97%, temperature 37.8°C. He is feeding well. How should his breathing be classified?

A Fast breathing, because any RR above 40 is fast
B Not fast breathing for his age (cut-off ≥50/min at 2–12 months)
C Severe respiratory distress
D Fast breathing, because he has fever
E Cannot be classified without a chest X-ray

2. A 2-year-old girl with diarrhoea is lethargic. Her hands are cold, capillary refill is 4 s, pulse 170/min and weak, BP 82/50 mmHg. MUAC 13.6 cm, no oedema, Hb 10 g/dL. What is the MOST accurate interpretation?

A She has shock (cold hands, CRT >3 s, weak fast pulse) and needs immediate IV fluid resuscitation
B She is not in shock because her BP is normal for age
C She has some dehydration and needs ORS 75 mL/kg over 4 hours
D She has SAM and needs ReSoMal
E She needs a CT head for lethargy

3. A 16-month-old boy is measured standing against a wall chart at a health centre: 74 cm. His weight is 8.0 kg. What is the MOST important correction before plotting?

A Measure him lying on a length board (or add 0.7 cm), because children under 2 years are measured lying
B No correction is needed
C Subtract 0.7 cm because he was standing
D Use weight-for-age only
E Measure his head circumference instead of length

4. A 22-month-old girl weighs 7.6 kg; length 77 cm; MUAC 11.3 cm; no oedema. She is alert, has no danger signs, and finishes the RUTF appetite test portion. What is the MOST appropriate classification and management?

A Moderate acute malnutrition: counselling only
B Uncomplicated severe acute malnutrition: outpatient therapeutic care with RUTF, amoxicillin and weekly follow-up
C Complicated SAM: admit for F-75
D Normal nutrition because WFA is only mildly low
E Stunting only: no treatment needed

5. A 2-year-old boy has swelling of both feet and lower legs that pits after 3 seconds of pressure, sparse reddish hair and peeling skin on the thighs. MUAC 12.9 cm; WFH Z-score −1.2. Urine dipstick: protein trace. What is the MOST likely classification?

A Normal nutrition because MUAC and WFH are normal
B Nephrotic syndrome
C Severe acute malnutrition (oedematous, kwashiorkor) with oedema grade ++
D Moderate acute malnutrition
E Heart failure

6. A 15-month-old girl with SAM (WFL < −3, MUAC 10.8 cm, no oedema) has had diarrhoea for 3 days. Her eyes look sunken and her skin pinch returns slowly, but she is alert and drinks eagerly. The mother says her eyes have looked like this for weeks. Which statement is MOST accurate? Advanced

A She has severe dehydration and needs Plan C Ringer's lactate 100 mL/kg
B Give 20 mL/kg normal saline bolus
C She has no dehydration because she is alert
D Sunken eyes and slow skin pinch are unreliable in SAM; treat cautiously as some dehydration with ReSoMal and monitor closely
E Withhold all fluids until electrolytes are available

7. A 3-day-old term baby (3.1 kg) is breastfeeding well. In daylight, after blanching, the skin is yellow down to the knees and on the lower arms, but not on the palms or soles. What is the MOST appropriate action?

A Reassure: jaundice is normal on day 3
B Refer for exchange transfusion immediately
C Stop breastfeeding and give glucose water
D Start phenobarbital
E Measure the bilirubin (TcB or serum) and plot it by hours of age on the nomogram

8. An 8-month-old boy has fever and vomiting. He is irritable, with a high-pitched cry. When held calmly upright, his anterior fontanelle feels tense and bulging. Which is the MOST important next step?

A Reassure: the fontanelle bulges when infants cry
B Arrange a skull X-ray
C Give ORS for dehydration
D Treat as possible meningitis: start antibiotics and assess for lumbar puncture
E Measure head circumference and review in 1 week

9. A 6-year-old girl's BP is measured with an infant cuff that covers only half the length of her upper arm. The reading is 132/88 mmHg. What is the MOST appropriate next step?

A Diagnose hypertension and start amlodipine
B Order a renal ultrasound immediately
C Ignore the reading because children are never hypertensive
D Check BP on the leg only
E Repeat with a correctly sized cuff (bladder width ≈40% of arm circumference) when she is calm

10. A 3-year-old boy (11.2 kg, height 88 cm) has WFH Z-score −1.4 and HFA Z-score −2.6. He is active and well. Which is the BEST interpretation?

A Severe wasting
B Moderate stunting without wasting: chronic undernutrition
C Normal growth
D Severe stunting with wasting
E Overweight

11. A 9-month-old girl is pale. Her palms are much paler than the examiner's and almost white. HR 168/min, RR 56/min, liver 4 cm below the costal margin, gallop rhythm. What is the MOST appropriate next step?

A Urgent Hb (HemoCue), malaria test, and prepare for transfusion; treat heart failure carefully
B Start oral iron and review in 1 month
C Give 20 mL/kg Ringer's lactate bolus
D Send stool for ova and cysts and wait
E Reassure: pallor is normal in infants

12. A 7-month-old girl is weighed fully clothed with a thick sweater, a nappy and shoes on a hanging scale: 6.5 kg. Her previous weight on the card 1 month ago (undressed) was 6.4 kg. What is the BEST conclusion?

A She gained 100 g in a month, which is adequate
B She has SAM
C The measurement is not comparable; reweigh her undressed before judging weight gain
D Weight is unreliable in infants; use MUAC only
E She has gained weight too quickly

13. A 5-year-old boy with a 3-week cough has firm, matted, non-tender left cervical lymph nodes 3 cm in size, and small (0.5 cm) mobile nodes in both groins. Which finding is MOST significant?

A The small inguinal nodes
B Neither: cough is the only important finding
C Both are normal for age
D The matted 3 cm cervical lymph nodes
E The absence of tender nodes means infection is excluded

14. At a growth-monitoring clinic, a 6-month-old's weight has fallen from the 0 Z-score line at 2 months to the −2 line now. Length is tracking along −0.5. What does this pattern indicate?

A Normal growth because she is still above −3
B Measurement error, so ignore it
C Stunting
D Overweight
E Growth faltering (crossing two major lines downward) needing assessment of feeding and illness

15. A 12-month-old boy has a head circumference of 50.5 cm (above +3 SD); 3 months ago it was at +1 SD. He has a tense fontanelle and his eyes look downwards ('setting sun'). Which is the MOST appropriate next step?

A Reassure: big heads run in families
B Recheck head circumference in 6 months
C Urgent cranial ultrasound (open fontanelle) or CT and neurosurgical referral for suspected hydrocephalus
D Start phenobarbital
E Order a skull X-ray only

16. A 4-month-old exclusively breastfed infant weighs 4.3 kg (weight-for-age Z-score −2.6) with MUAC 10.6 cm and no oedema. She is alert, breastfeeding, with no danger signs. According to WHO 2023, how is she classified? Advanced

A An infant under 6 months at risk of poor growth and development (WAZ < −2 and MUAC <11.0 cm), needing assessment of breastfeeding, the mother's health and illness
B Normal, because MUAC cut-offs apply only from 6 months
C Moderate acute malnutrition needing supplementary RUTF
D Uncomplicated SAM managed with RUTF at home
E Stunting only

Answers & explanations

> **1. B** · Not fast breathing for his age (cut-off ≥50/min at 2–12 months)
>
> **B** is correct: the WHO cut-off for fast breathing at **2–12 months is ≥50/min**; 46/min is normal, so this is a cough or cold without pneumonia (no indrawing, SpO_2 normal). **A** is wrong: ≥40/min is the cut-off for 1–5 years. **C** is wrong: there are no signs of distress. **D** is wrong: fever raises RR slightly, but the classification still uses age cut-offs. **E** is wrong: the classification is clinical.

2. A · She has shock (cold hands, CRT >3 s, weak fast pulse) and needs immediate IV fluid resuscitation

A is correct: the WHO definition of **shock** is **cold hands + CRT >3 s + weak fast pulse**; **BP is maintained until late** in children (82 mmHg is above 70 + 2 × 2 = 74). She needs rapid **Ringer's lactate** (20 mL/kg UCG; WHO 2016 10–20 mL/kg) and reassessment, then Plan C. **B** is wrong: normal BP does not exclude shock. **C** is wrong: lethargy and shock mean severe dehydration, not some. **D** is wrong: she is not malnourished (MUAC 13.6 cm, no oedema). **E** is wrong: lethargy here is from hypoperfusion.

3. A · Measure him lying on a length board (or add 0.7 cm), because children under 2 years are measured lying

A is correct: WHO standards use **recumbent length under 2 years**; ideally remeasure on a length board, or **add 0.7 cm** to a standing measurement. **B** is wrong: standing height under-reads compared with length and may misclassify him. **C** is wrong: 0.7 cm is subtracted when a child **≥2 years** is measured lying. **D** is wrong: WFA cannot separate wasting from stunting. **E** is wrong: HC does not replace length.

4. B · Uncomplicated severe acute malnutrition: outpatient therapeutic care with RUTF, amoxicillin and weekly follow-up

B is correct: **MUAC <11.5 cm = SAM**; she passes the **appetite test** and has **no complications**, so she is managed in **outpatient therapeutic care** (RUTF 150–185 kcal/kg/day per WHO 2023, amoxicillin per UCG, regular follow-up, malaria test, HIV test, vaccines). **A** is wrong: MUAC 11.3 cm is below the MAM range (11.5–<12.5). **C** is wrong: no complications and good appetite. **D** is wrong: MUAC alone classifies her as SAM. **E** is wrong: she is wasted, not only stunted.

5. C · Severe acute malnutrition (oedematous, kwashiorkor) with oedema grade ++

C is correct: **bilateral pitting oedema** of feet and lower legs (**++**) with hair and skin changes and little proteinuria is **kwashiorkor = SAM**, regardless of MUAC and WFH (oedema adds weight and can mask wasting). **A** is wrong: oedema overrides the numbers. **B** is wrong: nephrotic syndrome has heavy proteinuria (3+) and periorbital onset. **D** is wrong: any bilateral nutritional oedema is SAM. **E** is wrong: there are no cardiac signs described.

6. D · Sunken eyes and slow skin pinch are unreliable in SAM; treat cautiously as some dehydration with ReSoMal and monitor closely

D is correct: in **SAM**, wasting causes **sunken eyes and reduced skin turgor**; rely on the history (diarrhoea, thirst, recent change) and give **ReSoMal** (5 mL/kg every 30 min for 2 h, then 5–10 mL/kg/h alternating with F-75), watching for overload. **A** and **B** are wrong: standard Plan C volumes and boluses can cause heart failure in SAM, and she is not shocked. **C** is wrong: she has fluid losses and is thirsty. **E** is wrong: oral rehydration should not be delayed.

7. E · Measure the bilirubin (TcB or serum) and plot it by hours of age on the nomogram

E is correct: jaundice reaching **Kramer zone 4** suggests a significant level, and visual estimates are unreliable: **measure** and plot on an **hour-specific** nomogram (AAP 2022) to decide on phototherapy. **A** is wrong: physiological jaundice can still need phototherapy. **C** is wrong: stopping breastfeeding worsens jaundice; glucose water is not recommended. **D** is wrong: phenobarbital is not used to treat neonatal jaundice. **B** is wrong: exchange is decided on a measured level.

8. D · Treat as possible meningitis: start antibiotics and assess for lumbar puncture

D is correct: a **bulging fontanelle in a calm, upright febrile infant** suggests raised intracranial pressure from **meningitis** until proven otherwise: give antibiotics promptly (ceftriaxone 100 mg/kg) and assess for LP (contraindications include signs of raised ICP: senior decision). **A** is wrong: the assessment was done when calm and upright. **C** is wrong: dehydration causes a **sunken** fontanelle. **B** is wrong: X-ray does not help. **E** is wrong: delay is dangerous.

9. E · Repeat with a correctly sized cuff (bladder width ≈40% of arm circumference) when she is calm

E is correct: a **cuff that is too small over-reads** BP; repeat with the correct size (bladder width ~40% of arm circumference, length 80–100%) with the child calm and seated, and interpret against age/sex/height centiles. **A** is wrong: the reading is unreliable. **C** is wrong: children do get hypertension (glomerulonephritis, renal disease, coarctation). **D** is wrong: leg BP is for coarctation screening, not a replacement. **B** is wrong: investigate only after confirming true hypertension.

10. B · Moderate stunting without wasting: chronic undernutrition

B is correct: **HFA between −2 and −3 = moderate stunting** (chronic undernutrition); **WFH −1.4** means he is **not wasted**. Address diet quality, hygiene and chronic illness, and monitor height velocity. **A** and **D** are wrong: WFH is above −2. **C** is wrong: HFA below −2 is stunting. **E** is wrong: overweight is WFH above +2.

11. A · Urgent Hb (HemoCue), malaria test, and prepare for transfusion; treat heart failure carefully

A is correct: **severe palmar pallor** with **tachycardia, tachypnoea, hepatomegaly and gallop** suggests **decompensated severe anaemia** (heart failure): measure Hb now, test for malaria and **transfuse** (packed cells 10 mL/kg slowly or whole blood per guideline; furosemide if overloaded). **B** is wrong: oral iron is far too slow. **C** is wrong: crystalloid worsens heart failure and dilutes Hb. **D** is wrong: stool tests do not change urgent care. **E** is wrong: severe pallor is never normal.

12. C · The measurement is not comparable; reweigh her undressed before judging weight gain

C is correct: clothes and a wet nappy can weigh **0.3–0.5 kg** (up to 7% of her weight); the true weight may have fallen. **Reweigh undressed** on a calibrated scale before interpreting the trend. **A** is wrong: the gain may be artefactual (and 100 g/month is inadequate at 7 months anyway, normal ~10 g/day). **B** is wrong: SAM needs WFL, MUAC or oedema. **D** is wrong: weight is essential when measured correctly. **E** is wrong: no evidence supports that.

13. D · The matted 3 cm cervical lymph nodes

D is correct: **matted, firm cervical nodes >2 cm** with a chronic cough suggest **TB lymphadenitis** (also lymphoma, HIV): investigate with fine-needle aspirate for **Xpert Ultra**, CXR and HIV test. **A** is wrong: small mobile inguinal nodes (<1.5 cm) are common in children. **C** is wrong: 3 cm matted nodes are abnormal. **B** is wrong: the nodes are a key clue. **E** is wrong: TB nodes are classically non-tender.

14. E · Growth faltering (crossing two major lines downward) needing assessment of feeding and illness

E is correct: **crossing two major Z-score lines downward** is **growth faltering**; at 6 months think of feeding problems (end of exclusive breastfeeding, poor complementary feeding), infection (HIV, TB, diarrhoea) and the mother's health. **A** is wrong: the trend matters more than a single point. **C** is wrong: length is tracking normally, so this is weight faltering, not stunting. **D** is wrong: she is losing ground. **B** is wrong: assume it is real until a careful reweigh says otherwise.

15. C · Urgent cranial ultrasound (open fontanelle) or CT and neurosurgical referral for suspected hydrocephalus

C is correct: **rapidly crossing centiles upward**, a **tense fontanelle** and **setting-sun eyes** indicate **hydrocephalus**: cranial ultrasound or CT and **urgent neurosurgical referral** (e.g. CURE Children's Hospital, Mbale). **A** is wrong: familial macrocephaly grows parallel to centiles without raised ICP signs. **B** is wrong: delay risks brain damage. **D** is wrong: no seizures are described. **E** is wrong: X-ray adds little.

16. A · An infant under 6 months at risk of poor growth and development (WAZ < −2 and MUAC <11.0 cm), needing assessment of breastfeeding, the mother's health and illness

A is correct: the **WHO 2023 wasting guideline** defines infants **<6 months at risk of poor growth and development** by **WAZ < −2**, WLZ < −2, **MUAC <11.0 cm (from 6 weeks)** or oedema; management centres on **breastfeeding support**, the mother's physical and mental health, and treating illness, with inpatient care if severe or complicated. **B** is wrong: WHO now provides an infant MUAC cut-off. **C** and **D** are wrong: RUTF is not for infants under 6 months. **E** is wrong: stunting is defined by length-for-age, which is not given.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Measuring and classifying a wasted toddler

A 14-month-old girl is brought to a HC IV in Lira with diarrhoea for 5 days and poor appetite. Her mother stopped breastfeeding at 8 months when she became pregnant again. The child looks thin and irritable.

1. What do you do first, before measuring?

> **ETAT triage**: check for emergency signs (airway, breathing, shock: cold hands, CRT, pulse; coma; convulsions; severe dehydration). Then priority signs (visible severe wasting, oedema of both feet). Count **RR**, check **temperature** (hypothermia), and **glucose** if lethargic.

2. Describe how you weigh and measure her.

> Weigh **undressed** on a calibrated infant scale (record to 0.01–0.1 kg): **6.1 kg**. Measure **length lying** with two people on a length board: **71.0 cm**. **MUAC** on the left arm at the acromion–olecranon midpoint: **10.9 cm**. Check **bilateral pitting oedema** on the feet (3 s): none. Head circumference 43.5 cm.

3. Plot and interpret the Z-scores.

> **WFL < −3** (severe wasting), **LFA about −2** (borderline stunting), **WFA < −3** (severely underweight); **MUAC 10.9 cm = red**. **Diagnosis: SAM** (severe wasting).

4. How do you assess dehydration and what is the trap?

> She is irritable, drinks eagerly, eyes look sunken, skin pinch slow. In **SAM** these signs are unreliable (wasting itself causes them); use history (5 days of diarrhoea, mother says eyes recently sunken), thirst and urine. Treat as **some dehydration with ReSoMal** cautiously, not Plan C.

5. Is this complicated or uncomplicated SAM, and what is the next management step?

> Do the **appetite test** and look for complications. She refuses RUTF and has diarrhoea with dehydration → **complicated SAM: admit** for stabilisation: prevent hypoglycaemia and hypothermia, **ReSoMal**, **antibiotics** (ampicillin + gentamicin), **F-75 130 mL/kg/day** in 8–12 feeds, HIV test, malaria test, then transition to RUTF.

Take-away: Accurate weight, length, MUAC and an oedema check classify SAM; dehydration signs are unreliable in wasted children, and the appetite test decides inpatient versus outpatient care.

Case 2: The swollen child: kwashiorkor or nephrotic syndrome?

A 3-year-old boy is brought to Mengo with swelling that started around the eyes 10 days ago and now involves the legs and scrotum. He weighs 16.5 kg (up from 14.8 kg at a clinic visit 2 months ago), height 95 cm. He eats family meals including beans and fish. Skin and hair look normal.

1. What do you check on general examination?

> **Vital signs** (BP for nephritic/hypertension, HR, RR), **oedema distribution and grade** (periorbital, legs, scrotal, ascites; pitting), **skin and hair** (kwashiorkor changes absent), pallor, signs of infection (peritonitis, cellulitis), **pleural effusions**, heart (gallop, raised JVP), liver.

2. How do you interpret his weight and nutritional status?

> His weight has **risen 1.7 kg because of fluid**; WFH and MUAC overestimate nutrition when oedema is present. Oedema starting **periorbitally** in a well-fed child is typical of **nephrotic syndrome**, not kwashiorkor.

3. Which bedside test is most useful next?

> **Urine dipstick**: **protein 3+** supports nephrotic syndrome (also look for blood: nephritic features). Then urine protein:creatinine ratio, **serum albumin** (<25 g/L), cholesterol, creatinine, FBC; screen for **TB, HIV, hepatitis B and malaria** before steroids, and treat worms.

4. Dipstick shows protein 4+, no blood; BP 98/60. What is your plan?

> **Idiopathic nephrotic syndrome** (likely minimal change): **prednisolone 60 mg/m²/day (2 mg/kg, max 60 mg)** for 4 weeks then alternate-day tapering (KDIGO 2021), **salt restriction**, daily weight, urine dipstick monitoring, treat infections (peritonitis risk), avoid excessive diuretics (thrombosis/hypovolaemia), vaccinate (pneumococcal) when stable.

5. How would a child with kwashiorkor have differed?

> Poor diet history, oedema starting in the **feet**, **flaky-paint dermatosis**, **sparse pale hair**, apathy, hepatomegaly, little or trace proteinuria; managed as **SAM** (F-75, antibiotics, no diuretics).

Take-away: Bilateral oedema needs its distribution, a urine dipstick and a diet history: heavy proteinuria and periorbital onset point to nephrotic syndrome, skin/hair changes and poor diet to kwashiorkor.

Case 3: A jaundiced newborn on the postnatal ward

A 40-hour-old baby girl born at 37 weeks (birth weight 2.9 kg) is noticed to be yellow by the midwife at Mengo. Her mother is blood group O Rh-positive. The baby is breastfeeding 6 times a day. Temperature 36.8°C, HR 140/min, RR 48/min.

1. How do you examine for jaundice?

> In **daylight** (or good white light), **blanch** the skin with a finger over the forehead, chest, abdomen, knees, arms and **palms and soles**; look at sclerae and gums. Note the furthest zone (Kramer). Here yellow reaches the **lower abdomen and thighs (zone 3)**.

2. What else do you examine in the general examination?

> **Weight** (loss since birth: 2.68 kg = 7.6% loss), **hydration** (fontanelle, mucous membranes, urine/stool output), **tone, alertness, suck and cry** (encephalopathy signs), **pallor and hepatosplenomegaly** (haemolysis), signs of **sepsis** (temperature, lethargy, RR), cephalhaematoma or bruising.

3. What do you do with your visual estimate?

> **Measure**: TcB, then **total serum bilirubin** if near the threshold; plot by **hours of age** on the AAP 2022 nomogram for 37 weeks, noting **risk factors** (37 weeks, possible **ABO incompatibility** with an O mother, need for G6PD testing).

4. Total serum bilirubin is above the phototherapy line. What is your plan?

> **Phototherapy** (eyes covered, maximal skin exposure, turn regularly), **increase breastfeeding** to 8–12 times/day (express if needed), monitor temperature and hydration; check **blood groups, DAT, FBC/film, G6PD**; repeat bilirubin in 4–6 h; consider exchange transfusion if it rises towards the exchange line or encephalopathy signs appear.

5. Which findings would make you escalate urgently?

> Jaundice on **palms and soles**, a rapid rise, bilirubin near the **exchange threshold**, **lethargy, poor suck, high-pitched cry, hypertonia/arching, setting-sun eyes** (acute bilirubin encephalopathy), fever or other signs of sepsis, pallor (haemolysis).

Take-away: Neonatal jaundice is examined in good light with blanching, but decisions rest on a measured bilirubin plotted by hours of age and risk factors.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.